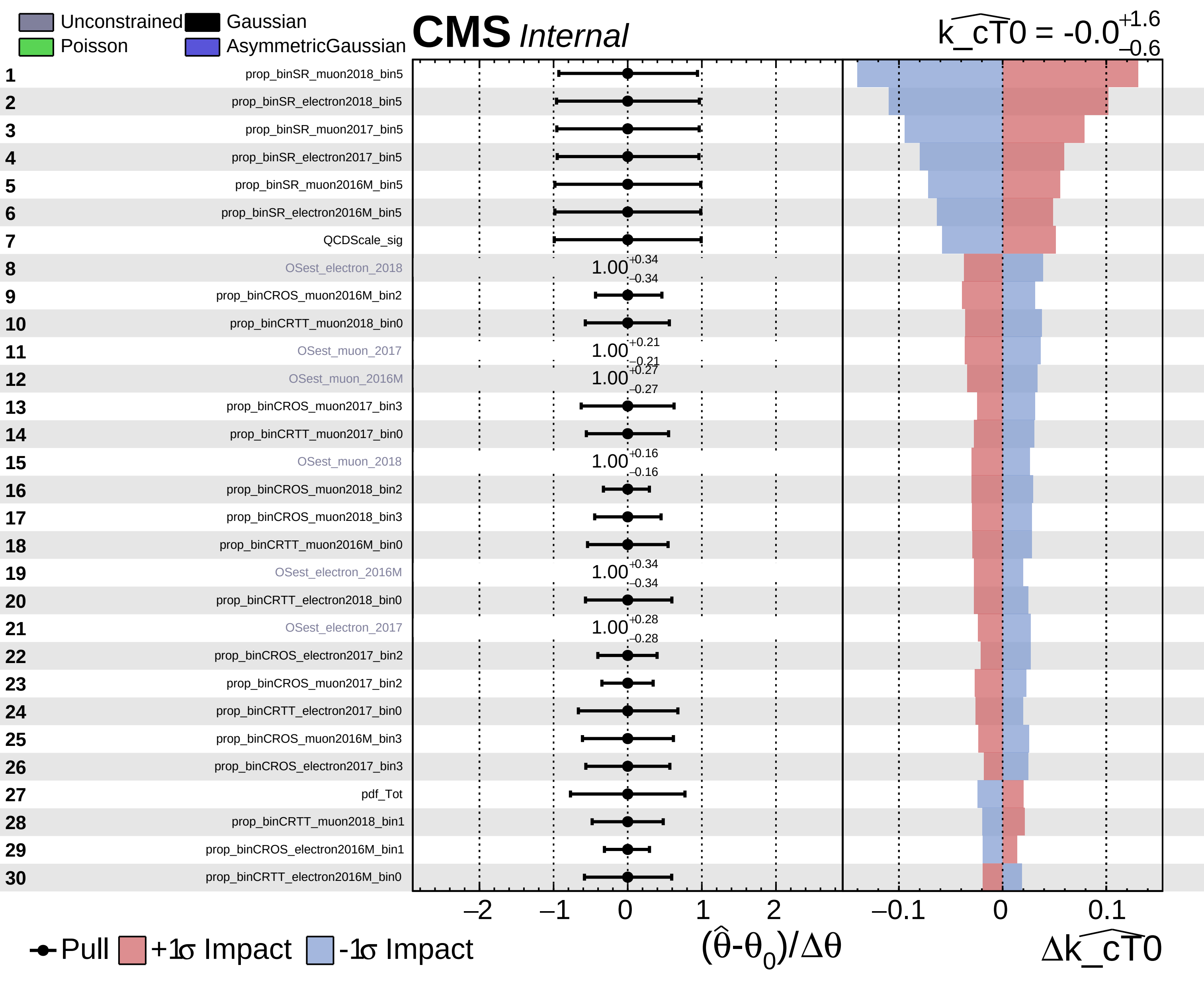
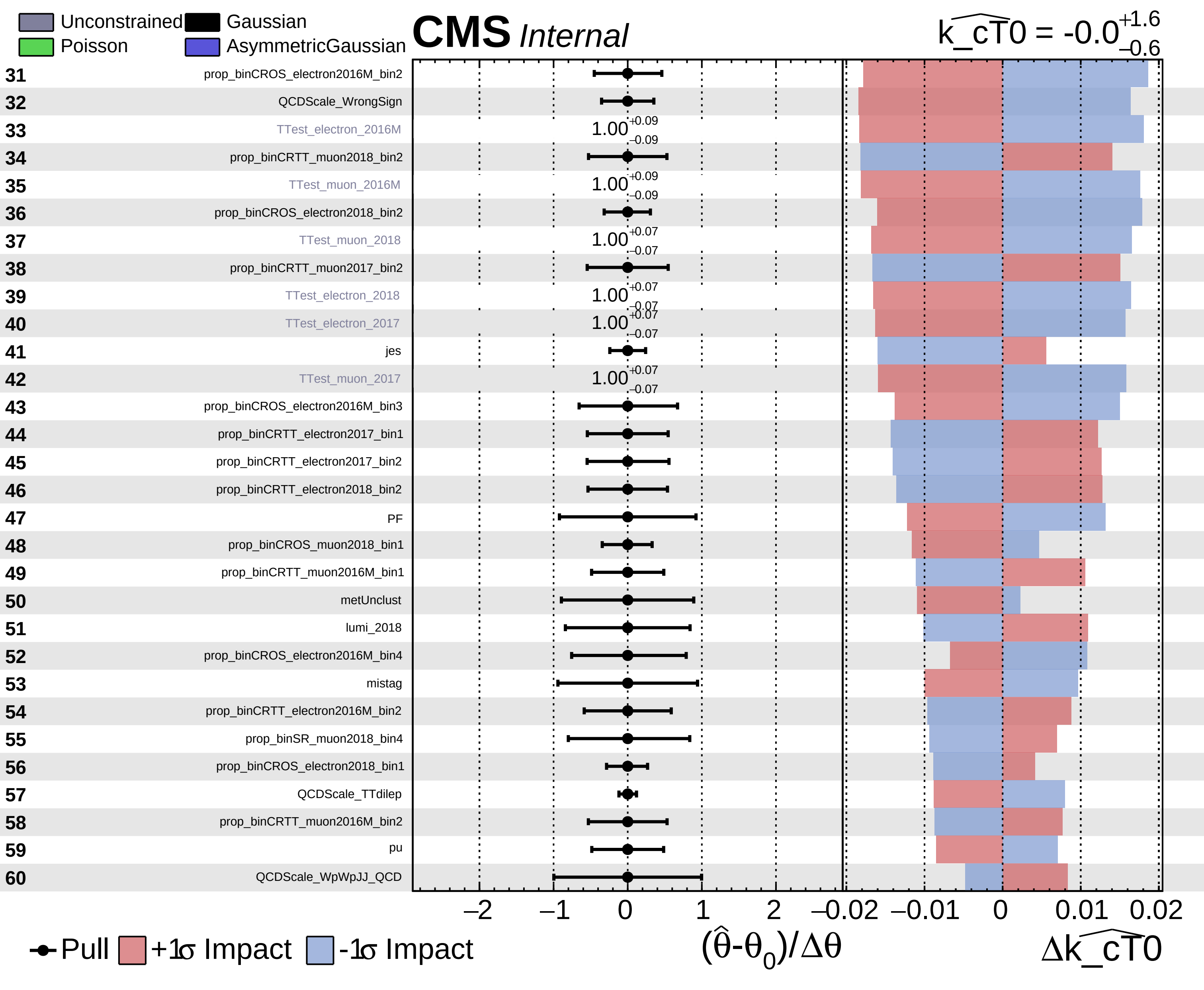
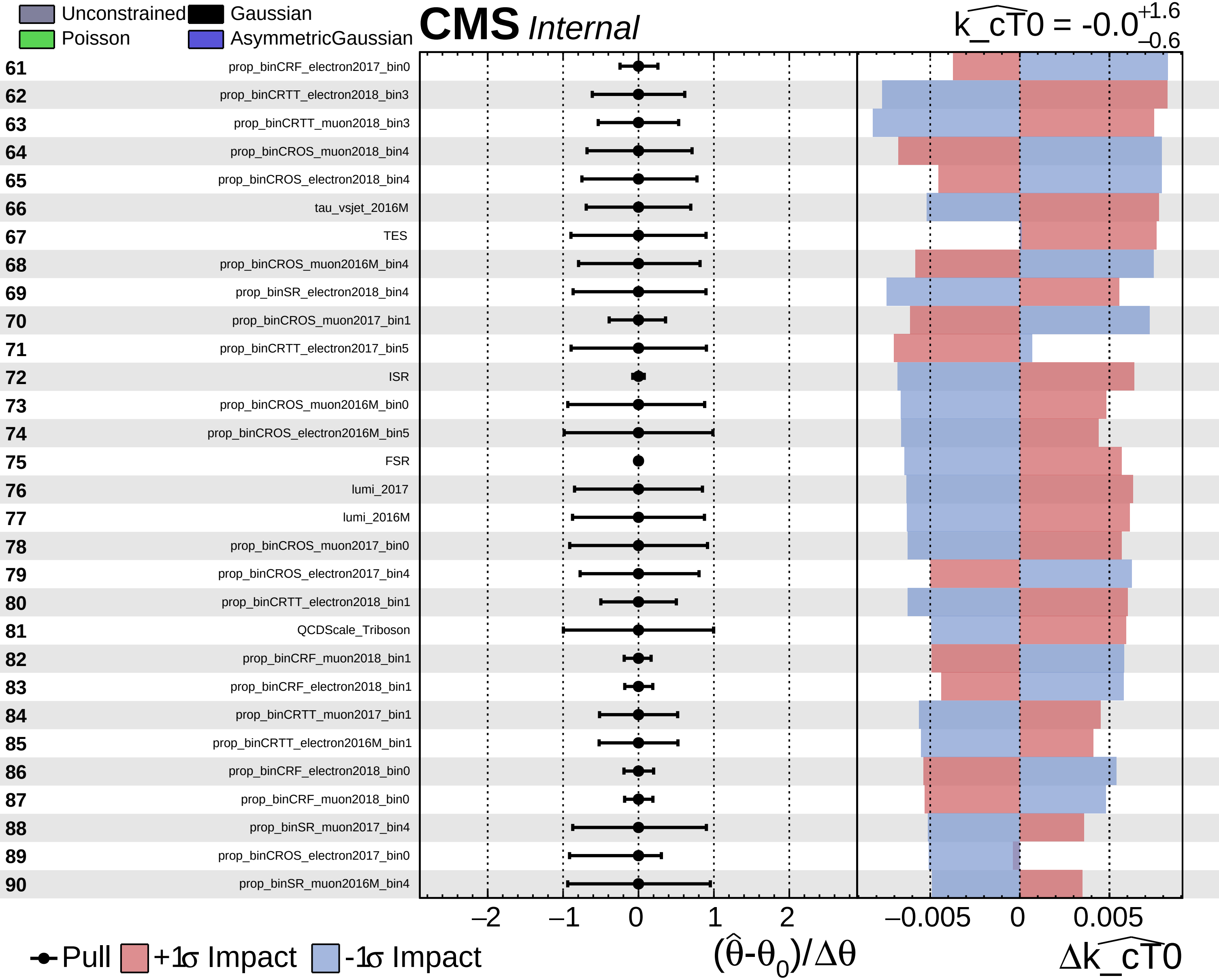
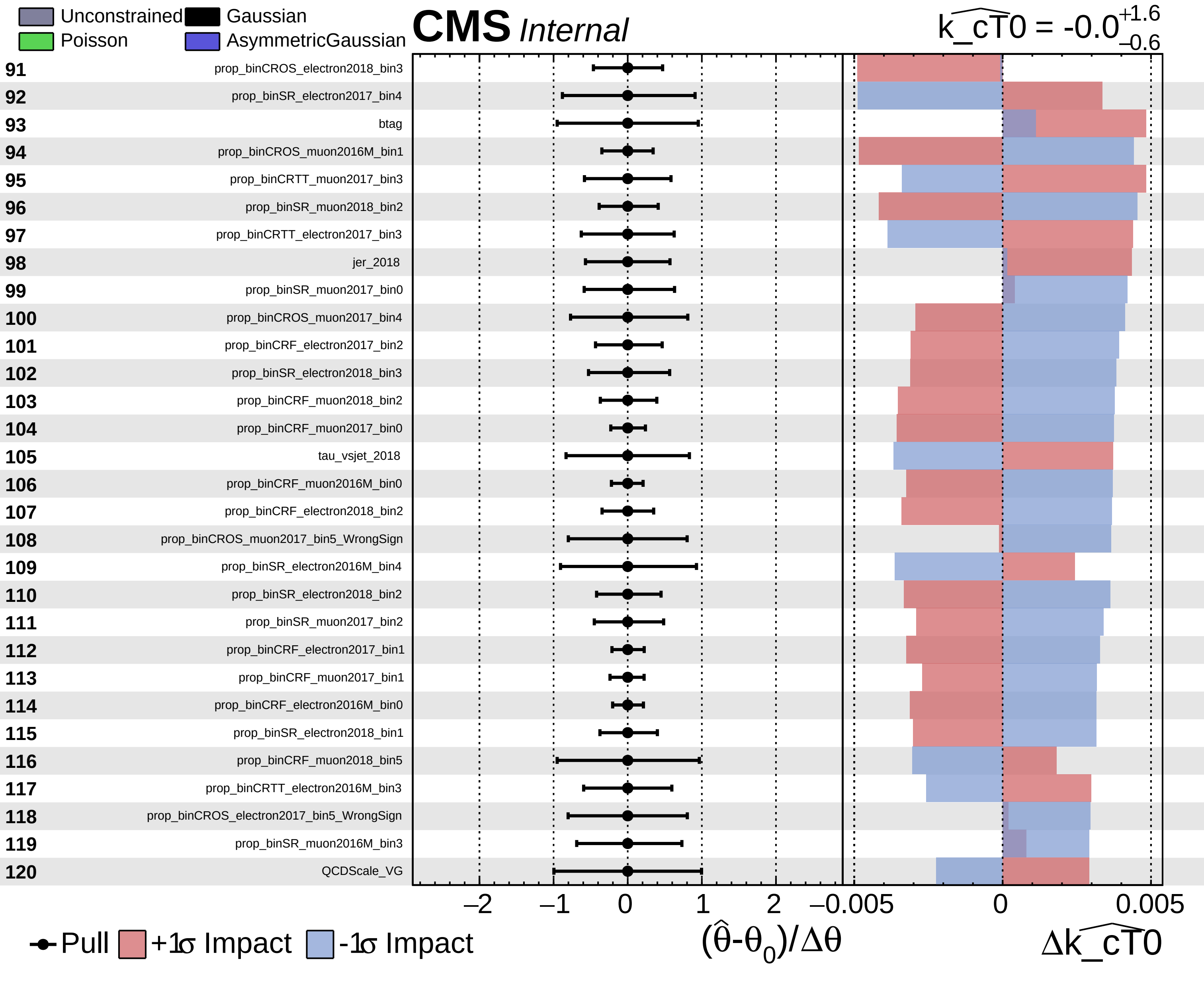


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.0^{+1.6}_{-0.6}$
1 prop_binSR_muon2018_bin5
2 prop_binSR_electron2018_bin5
3 prop_binSR_muon2017_bin5
4 prop_binSR_electron2017_bin5
5 prop_binSR_muon2016M_bin5
6 prop_binSR_electron2016M_bin5
7 QCDScale_sig
8 OSest_electron_2018 $1.00^{+0.34}_{-0.34}$
9 prop_binCROS_muon2016M_bin2
10 prop_binCRTT_muon2018_bin0
11 OSest_muon_2017 $1.00^{+0.21}_{-0.21}$
12 OSest_muon_2016M $1.00^{+0.27}_{-0.27}$
13 prop_binCROS_muon2017_bin3
14 prop_binCRTT_muon2017_bin0
15 OSest_muon_2018 $1.00^{+0.16}_{-0.16}$
16 prop_binCROS_muon2018_bin2
17 prop_binCROS_muon2018_bin3
18 prop_binCRTT_muon2016M_bin0
19 OSest_electron_2016M $1.00^{+0.34}_{-0.34}$
20 prop_binCRTT_electron2018_bin0
21 OSest_electron_2017 $1.00^{+0.28}_{-0.28}$
22 prop_binCROS_electron2017_bin2
23 prop_binCROS_muon2017_bin2
24 prop_binCRTT_electron2017_bin0
25 prop_binCROS_muon2016M_bin3
26 prop_binCROS_electron2017_bin3
27 pdf_Tot
28 prop_binCRTT_muon2018_bin1
29 prop_binCROS_electron2016M_bin1
30 prop_binCRTT_electron2016M_bin0
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

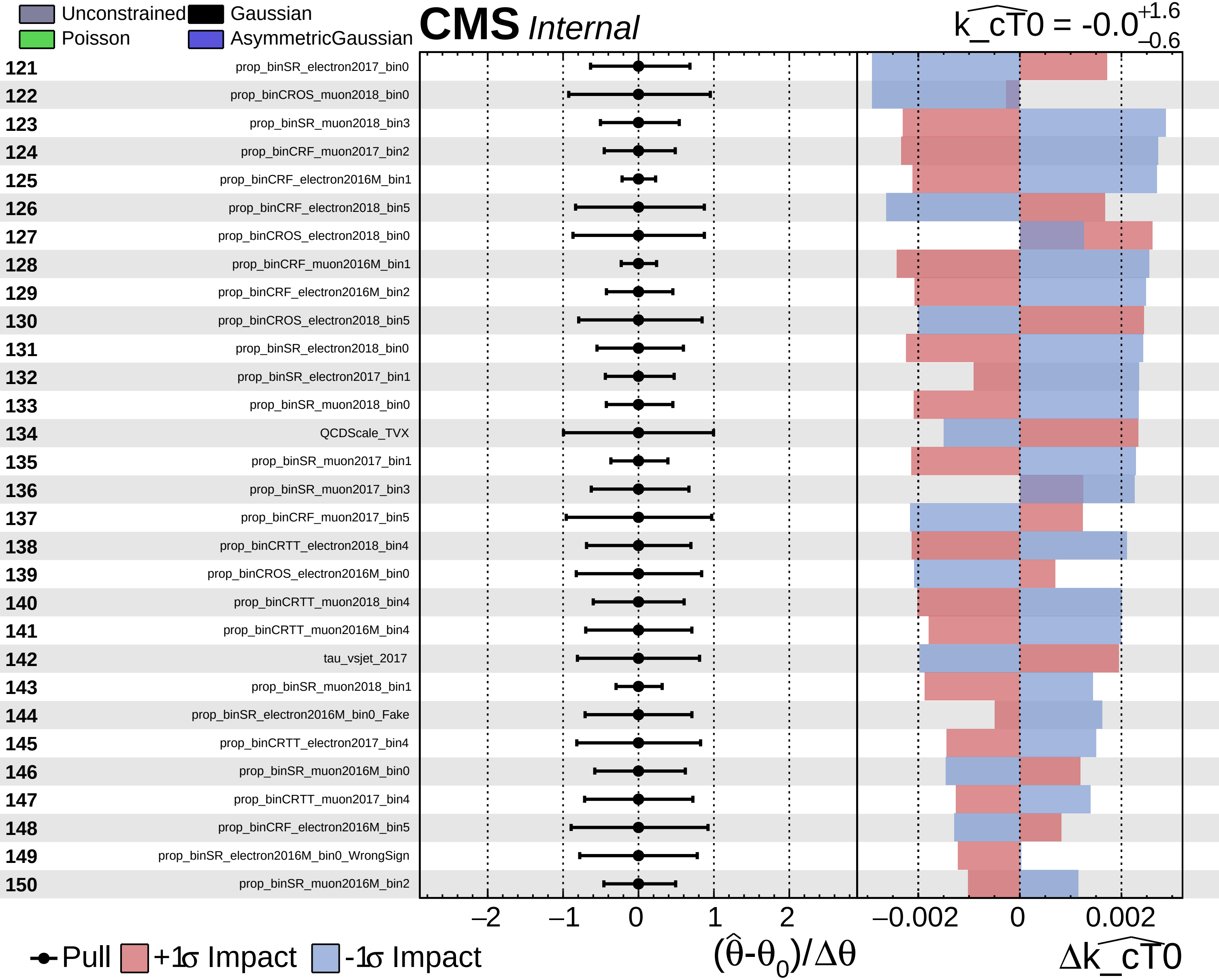


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.0 +1.6 −0.6
31 prop_binCROS_electron2016M_bin2
32 QCDScale_WrongSign
33 TTest_electron_2016M 1.00 +0.09 −0.09
34 prop_binCRTT_muon2018_bin2
35 TTest_muon_2016M 1.00 +0.09 −0.09
36 prop_binCROS_electron2018_bin2
37 TTest_muon_2018 1.00 +0.07 −0.07
38 prop_binCRTT_muon2017_bin2
39 TTest_electron_2018 1.00 +0.07 −0.07
40 TTest_electron_2017 1.00 +0.07 −0.07
41 jes
42 TTest_muon_2017 1.00 +0.07 −0.07
43 prop_binCROS_electron2016M_bin3
44 prop_binCRTT_electron2017_bin1
45 prop_binCRTT_electron2017_bin2
46 prop_binCRTT_electron2018_bin2
47 PF
48 prop_binCROS_muon2018_bin1
49 prop_binCRTT_muon2016M_bin1
50 metUnclust
51 lumi_2018
52 prop_binCROS_electron2016M_bin4
53 mistag
54 prop_binCRTT_electron2016M_bin2
55 prop_binSR_muon2018_bin4
56 prop_binCROS_electron2018_bin1
57 QCDScale_TTdilep
58 prop_binCRTT_muon2016M_bin2
59 pu
60 QCDScale_WpWpJJ_QCD
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT0

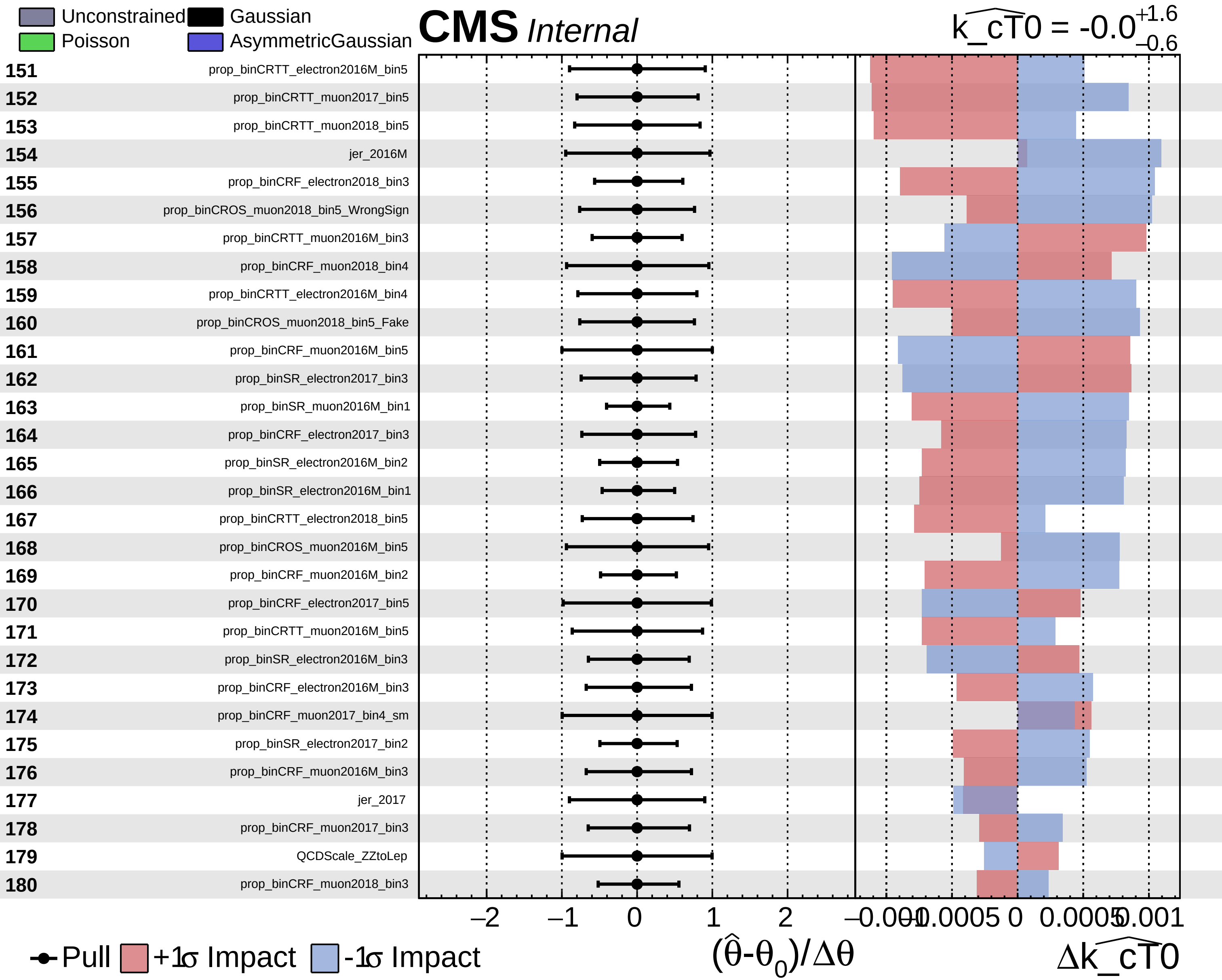

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.0^{+1.6}_{-0.6}$
61 prop_binCRF_electron2017_bin0
62 prop_binCRTT_electron2018_bin3
63 prop_binCRTT_muon2018_bin3
64 prop_binCROS_muon2018_bin4
65 prop_binCROS_electron2018_bin4
66 tau_vsjet_2016M
67 TES
68 prop_binCROS_muon2016M_bin4
69 prop_binSR_electron2018_bin4
70 prop_binCROS_muon2017_bin1
71 prop_binCRTT_electron2017_bin5
72 ISR
73 prop_binCROS_muon2016M_bin0
74 prop_binCROS_electron2016M_bin5
75 FSR
76 lumi_2017
77 lumi_2016M
78 prop_binCROS_muon2017_bin0
79 prop_binCROS_electron2017_bin4
80 prop_binCRTT_electron2018_bin1
81 QCDScale_Triboson
82 prop_binCRF_muon2018_bin1
83 prop_binCRF_electron2018_bin1
84 prop_binCRTT_muon2017_bin1
85 prop_binCRTT_electron2016M_bin1
86 prop_binCRF_electron2018_bin0
87 prop_binCRF_muon2018_bin0
88 prop_binSR_muon2017_bin4
89 prop_binCROS_electron2017_bin0
90 prop_binSR_muon2016M_bin4
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

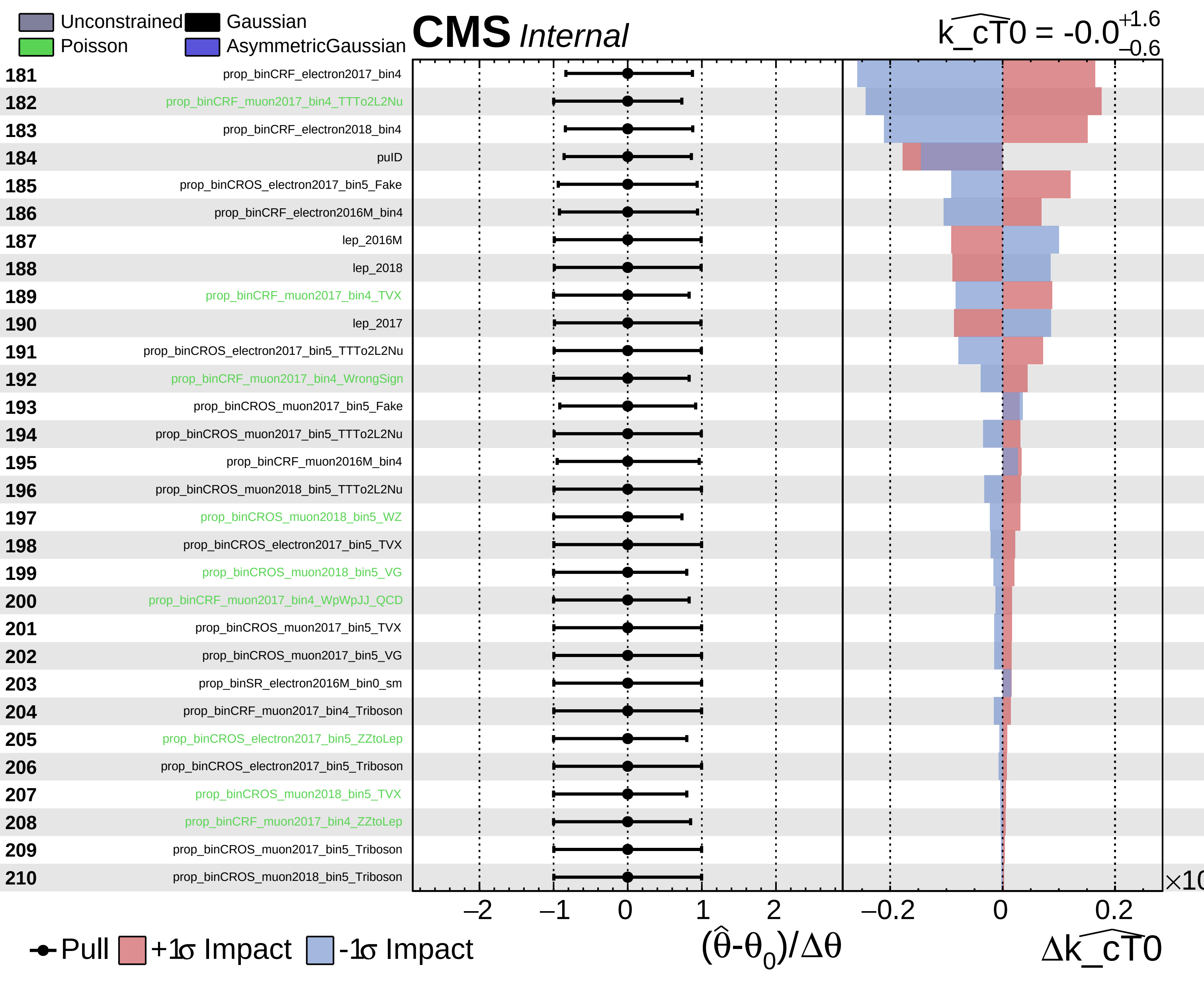


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.0^{+1.6}_{-0.6}$
91 prop_binCROS_electron2018_bin3
92 prop_binSR_electron2017_bin4
93 btag
94 prop_binCROS_muon2016M_bin1
95 prop_binCRTT_muon2017_bin3
96 prop_binSR_muon2018_bin2
97 prop_binCRTT_electron2017_bin3
98 jer_2018
99 prop_binSR_muon2017_bin0
100 prop_binCROS_muon2017_bin4
101 prop_binCRF_electron2017_bin2
102 prop_binSR_electron2018_bin3
103 prop_binCRF_muon2018_bin2
104 prop_binCRF_muon2017_bin0
105 tau_vsjet_2018
106 prop_binCRF_muon2016M_bin0
107 prop_binCRF_electron2018_bin2
108 prop_binCROS_muon2017_bin5_WrongSign
109 prop_binSR_electron2016M_bin4
110 prop_binSR_electron2018_bin2
111 prop_binSR_muon2017_bin2
112 prop_binCRF_electron2017_bin1
113 prop_binCRF_muon2017_bin1
114 prop_binCRF_electron2016M_bin0
115 prop_binSR_electron2018_bin1
116 prop_binCRF_muon2018_bin5
117 prop_binCRTT_electron2016M_bin3
118 prop_binCROS_electron2017_bin5_WrongSign
119 prop_binSR_muon2016M_bin3
120 QCDScale_VG
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

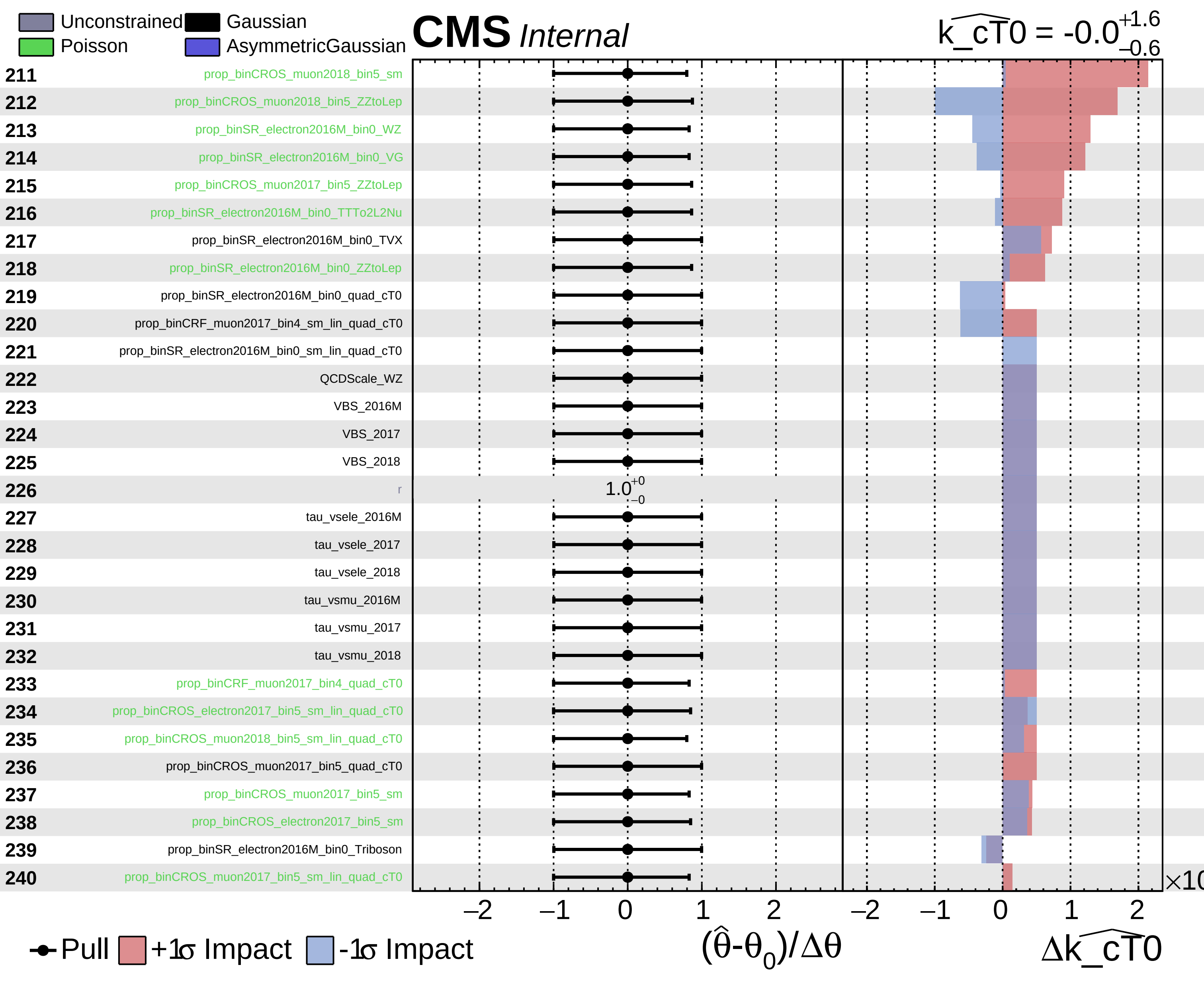


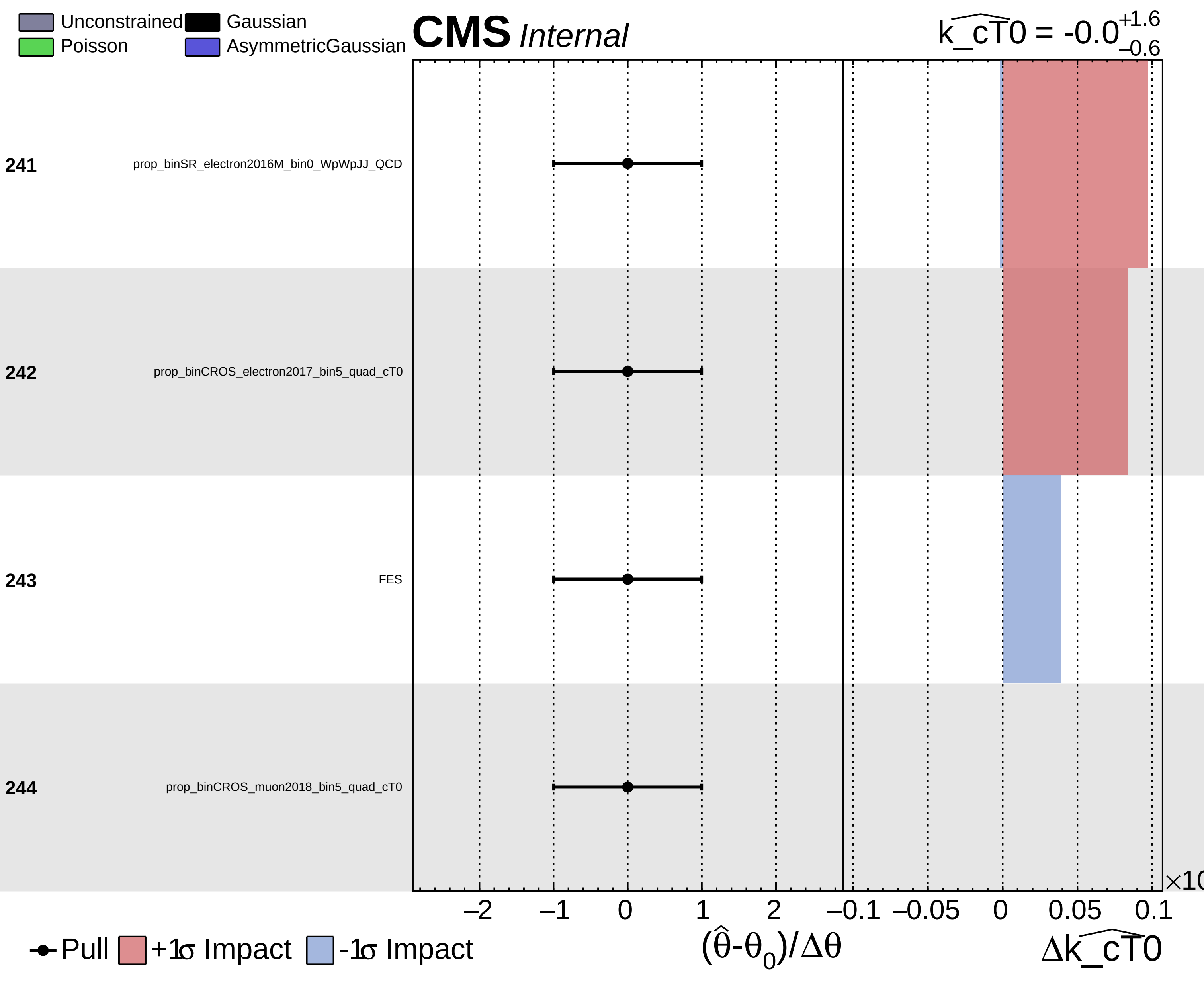
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.0+1.6−0.6
121 prop_binSR_electron2017_bin0
122 prop_binCROS_muon2018_bin0
123 prop_binSR_muon2018_bin3
124 prop_binCRF_muon2017_bin2
125 prop_binCRF_electron2016M_bin1
126 prop_binCRF_electron2018_bin5
127 prop_binCROS_electron2018_bin0
128 prop_binCRF_muon2016M_bin1
129 prop_binCRF_electron2016M_bin2
130 prop_binCROS_electron2018_bin5
131 prop_binSR_electron2018_bin0
132 prop_binSR_electron2017_bin1
133 prop_binSR_muon2018_bin0
134 QCDScale_TVX
135 prop_binSR_muon2017_bin1
136 prop_binSR_muon2017_bin3
137 prop_binCRF_muon2017_bin5
138 prop_binCRTT_electron2018_bin4
139 prop_binCROS_electron2016M_bin0
140 prop_binCRTT_muon2018_bin4
141 prop_binCRTT_muon2016M_bin4
142 tau_vsjet_2017
143 prop_binSR_muon2018_bin1
144 prop_binSR_electron2016M_bin0_Fake
145 prop_binCRTT_electron2017_bin4
146 prop_binSR_muon2016M_bin0
147 prop_binCRTT_muon2017_bin4
148 prop_binCRF_electron2016M_bin5
149 prop_binSR_electron2016M_bin0_WrongSign
150 prop_binSR_muon2016M_bin2
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.0 +1.6 −0.6
151 prop_binCRTT_electron2016M_bin5
152 prop_binCRTT_muon2017_bin5
153 prop_binCRTT_muon2018_bin5
154 jer_2016M
155 prop_binCRF_electron2018_bin3
156 prop_binCROS_muon2018_bin5_WrongSign
157 prop_binCRTT_muon2016M_bin3
158 prop_binCRF_muon2018_bin4
159 prop_binCRTT_electron2016M_bin4
160 prop_binCROS_muon2018_bin5_Fake
161 prop_binCRF_muon2016M_bin5
162 prop_binSR_electron2017_bin3
163 prop_binSR_muon2016M_bin1
164 prop_binCRF_electron2017_bin3
165 prop_binSR_electron2016M_bin2
166 prop_binSR_electron2016M_bin1
167 prop_binCRTT_electron2018_bin5
168 prop_binCROS_muon2016M_bin5
169 prop_binCRF_muon2016M_bin2
170 prop_binCRF_electron2017_bin5
171 prop_binCRTT_muon2016M_bin5
172 prop_binSR_electron2016M_bin3
173 prop_binCRF_electron2016M_bin3
174 prop_binCRF_muon2017_bin4_sm
175 prop_binSR_electron2017_bin2
176 prop_binCRF_muon2016M_bin3
177 jer_2017
178 prop_binCRF_muon2017_bin3
179 QCDScale_ZZtoLep
180 prop_binCRF_muon2018_bin3
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.0 +1.6 −0.6
181 prop_binCRF_electron2017_bin4
182 prop_binCRF_muon2017_bin4_TTTo2L2Nu
183 prop_binCRF_electron2018_bin4
184 puID
185 prop_binCROS_electron2017_bin5_Fake
186 prop_binCRF_electron2016M_bin4
187 lep_2016M
188 lep_2018
189 prop_binCRF_muon2017_bin4_TVX
190 lep_2017
191 prop_binCROS_electron2017_bin5_TTTo2L2Nu
192 prop_binCRF_muon2017_bin4_WrongSign
193 prop_binCROS_muon2017_bin5_Fake
194 prop_binCROS_muon2017_bin5_TTTo2L2Nu
195 prop_binCRF_muon2016M_bin4
196 prop_binCROS_muon2018_bin5_TTTo2L2Nu
197 prop_binCROS_muon2018_bin5_WZ
198 prop_binCROS_electron2017_bin5_TVX
199 prop_binCROS_muon2018_bin5_VG
200 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
201 prop_binCROS_muon2017_bin5_TVX
202 prop_binCROS_muon2017_bin5_VG
203 prop_binSR_electron2016M_bin0_sm
204 prop_binCRF_muon2017_bin4_Triboson
205 prop_binCROS_electron2017_bin5_ZZtoLep
206 prop_binCROS_electron2017_bin5_Triboson
207 prop_binCROS_muon2018_bin5_TVX
208 prop_binCRF_muon2017_bin4_ZZtoLep
209 prop_binCROS_muon2017_bin5_Triboson
210 prop_binCROS_muon2018_bin5_Triboson
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.0 +1.6 −0.6
211 prop_binCROS_muon2018_bin5_sm
212 prop_binCROS_muon2018_bin5_ZZtoLep
213 prop_binSR_electron2016M_bin0_WZ
214 prop_binSR_electron2016M_bin0_VG
215 prop_binCROS_muon2017_bin5_ZZtoLep
216 prop_binSR_electron2016M_bin0_TTTo2L2Nu
217 prop_binSR_electron2016M_bin0_TVX
218 prop_binSR_electron2016M_bin0_ZZtoLep
219 prop_binSR_electron2016M_bin0_quad_cT0
220 prop_binCRF_muon2017_bin4_sm_lin_quad_cT0
221 prop_binSR_electron2016M_bin0_sm_lin_quad_cT0
222 QCDScale_WZ
223 VBS_2016M
224 VBS_2017
225 VBS_2018
226 r 1.0 +0 −0
227 tau_vsele_2016M
228 tau_vsele_2017
229 tau_vsele_2018
230 tau_vsmu_2016M
231 tau_vsmu_2017
232 tau_vsmu_2018
233 prop_binCRF_muon2017_bin4_quad_cT0
234 prop_binCROS_electron2017_bin5_sm_lin_quad_cT0
235 prop_binCROS_muon2018_bin5_sm_lin_quad_cT0
236 prop_binCROS_muon2017_bin5_quad_cT0
237 prop_binCROS_muon2017_bin5_sm
238 prop_binCROS_electron2017_bin5_sm
239 prop_binSR_electron2016M_bin0_Triboson
240 prop_binCROS_muon2017_bin5_sm_lin_quad_cT0
−2 −1 0 1 2
−2 −1 0 1 2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.0^{+1.6}_{-0.6}$
241
prop_binSR_electron2016M_bin0_WpWpJJ_QCD
242
prop_binCROS_electron2017_bin5_quad_cT0
243
FES
244
prop_binCROS_muon2018_bin5_quad_cT0
−2
−1
0
1
2
−0.1
−0.05
0
0.05
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$